Notes 5

1. Linux Commands

`ls`

Definition: The `ls` command lists files and directories.

Usage: Displays the contents of the current directory by default. You can also specify a directory to list its contents.

Examples:

- ls
- ls -l
- ls -a

`pwd`

Definition: The `pwd` command stands for Print Working Directory. It displays the full path of the current directory.

Usage: Use `pwd` to find out which directory you are currently working in.

Examples:

- pwd

`cd`

Definition: The `cd` command stands for Change Directory. It allows you to move from one directory to another.

Usage: Use `cd` followed by a directory name or path to change your current working directory.

Examples:

- cd Documents
- cd ~
- cd ..

Questions and Answers

A. What is a variable?

A variable is a named place used to store a value that can be accessed or changed.

B. How do I use a variable?

In Bash, assign a value using an equals sign with no spaces around it. Use `$` followed by the variable name to access its value.

C. What is an environment variable?

An environment variable stores information used by the operating system and running programs. Examples include `HOME`, `PATH`, and `USER`.

D. What is a user-defined variable?

A user-defined variable is a variable created by a user or programmer to store information for a particular task.

E. What is the root directory?

The root directory is the top-level directory in a Linux filesystem. It is represented by `/`, and all other directories and files are located under it.

F. What does "Parent Directory" mean?

A parent directory is the directory directly above another directory in the filesystem hierarchy. The notation `..` represents the parent directory.

G. What does "Current Working Directory" mean?

The current working directory is the directory where commands are currently being executed.

H. What is an absolute path? Include an example.

An absolute path gives the complete location of a file or directory starting from the root directory (`/`). It does not depend on the current working directory.

Example: `/home/student/Documents/notes.txt`

I. What is a relative path? Include an example.

A relative path describes the location of a file or directory in relation to the current working directory. It does not begin with `/`.

Example: `Documents/notes.txt`

J. What is the difference between "Your home directory" and "The home directory"?

Your home directory is the personal directory assigned to your user account, such as `/home/student`.

The home directory can refer to a user's personal directory in general. The `/home` directory commonly contains the home directories of regular users on Linux systems. The `~` symbol represents your own home directory.